

LICONATE TABLETS 300MG

NAME AND STRENGTH OF ACTIVE INGREDIENT(S):

Each tablet contains:

Lithium Carbonate..... 300mg

PRODUCT DESCRIPTION:

A white, flat of diameter 11mm round tablet with 'MPI' logo.

PHARMACODYNAMICS:

Lithium possesses anti-manic effects but the mechanism is unknown; lithium ions may act by competing with sodium ions at various sites in the body.

PHARMACOKINETICS:

Lithium is readily and completely absorbed from the gastro-intestinal tract when taken as one of its salts. Absorption can be affected by the formulation of preparation taken. Peak serum concentrations are obtained between 0.5 and 3 hours after ingestion from conventional tablets. Lithium is distributed throughout the body and distribution is complete within 6 to 10 hours; higher concentration occur in the bones, the thyroid gland, and portions of the brain, than in the serum. Lithium is excreted predominantly in the urine; only a small amount can be detected in the faeces, saliva and sweat. It is not bound to plasma proteins. It crosses the placenta and is excreted in breast milk. In patients with normal renal function the average elimination half-life is about 20 to 24 hours.

INDICATION:

It is used in the prophylaxis and treatment of mania or hypomania, in the prophylaxis of manic depression (bipolar illness or bipolar depression) and recurrent depression (unipolar illness or unipolar depression).

RECOMMENDED DOSAGE:

To be take orally. This medication may be taken with food, to minimize stomach upset.

Acute treatment of mania and hypomania :

1.5 – 2g daily may be given in divided doses for the first 5 to 7 days, together with close checks on serum-lithium concentrations, a higher concentration, often up to 1.5 mmol/L may be required. The dose may need to be reduced rapidly once the acute phase has passed.

Prophylactic therapy :

The usual dose is 0.5 – 1.2g daily in divided doses. During the initial stages of lithium therapy, serum-lithium concentration must be checked at least once a week to ensure that they remain within the usual therapeutic range of 0.4 – 1.5mmol/L. Dosage adjustment should be done 5 to 7 days after the initiation of adjustment, according to the results of serum-lithium obtained. When consistent concentrations have been achieved, estimation may be made monthly and eventually, every 2 months.

However, dosage should be individualised according to clinical response to the therapy.

Intoxication may be expected at concentration exceeding 1.5 mmol/L although in some susceptible patients or the elderly, where intoxication may occur with concentration in excess of 1.0mmol/L.

Children : Not recommended for children under 12 years of age.

ROUTE OF ADMINISTRATION:

Oral

CONTRAINDICATIONS:

Severe renal disease, in debilitated or dehydrated patients, and in those with sodium depletion; and also in cardiovascular disease and hypothyroidism, and also in Addison's Disease. It should not be used in children under 12 years of age. Diuretic is normally contraindicated in patients receiving lithium therapy. Unless deemed essential, lithium should be avoided in patients with any significant cardiovascular or renal conditions.

WARNINGS AND PRECAUTIONS:

1. The margin between the therapeutic and toxic concentrations of lithium is narrow, therefore it should be given under close medical supervision, and serum concentration should be monitored regularly under controlled conditions.
2. Impaired driving performance or machinery operation skills may occur in patients receiving lithium.
3. It should be used with particular care in the elderly.
4. Patients should be examined periodically for abnormal thyroid function. Cardiac and renal function should be monitored regularly.

5. Doses must be taken at exactly the stipulated time of the day. Never compensate for an omitted dose by subsequently taking a double dose. Patients should not switch between different preparations without special control.
6. Reduction in sodium intake increases the amount of lithium retained by the kidney leading to a rise in serum-lithium concentrations, therefore, patients should avoid low salt dietary regimens or other dietary changes which may reduce sodium intake, or circumstances which may cause excessive sodium loss such as those leading to sweating, vomiting, or diarrhoea; with resultant dehydration. Conversely, increase sodium intake increases the amount of lithium excreted by kidneys, leading to reduced serum-lithium concentrations, therefore sodium containing medicaments, such as 'fruit salts', should not be given concomitantly with lithium salts.
7. Patients on long term maintenance lithium therapy do appear to be susceptible to the development of progressive impairment of urinary concentration ability but that it was most noticeable in patients with history of acute lithium toxicity.
8. Suicidality in Children and Adolescents
 - Antidepressants increase the risk of suicidal thinking and behavior (suicidality) in children and adolescents with major depressive disorder (MDD) and other psychiatric disorders.
 - Anyone considering the use of an antidepressant in a child or adolescent for any clinical use must balance the risk of increased suicidality with the clinical need.
 - Patients who are started on therapy should be observed closely for clinical worsening, suicidality, or unusual changes in behavior.
 - Families and caregivers should be advised to closely observe the patient and to communicate with the prescriber.
 - The indication(s) approved in paediatric for the particular drug should be clearly stated / included.

INTERACTIONS WITH OTHER MEDICAMENTS:

1. Lithium toxicity may be precipitated by concurrent administration of lithium with antihypertensive agents such as enalapril and methyldopa; antimicrobial agents such as metronidazole and spectinomycin; diuretics such as thiazide; and NSAIDs such as diclofenac, ibuprofen, indomethacin, naproxen, piroxicam, ketoprofen; and phenothiazine.
2. Blood lithium concentrations may be reduced by concomitant administration of lithium with sodium containing drug, gastrointestinal agents such as sodium bicarbonate and ispaghula, xanthines such as theophylline, and acetazolamide.
3. Neurotoxicity despite 'normal' serum-lithium concentrations has been described in patients taking concomitant lithium and phenytoin and/or phenobarbitone, or calcium-channel blockers such as verapamil and diltiazem, or neuroleptics such as flupenthixol decanoate, fluphenazine decanoate or high dose haloperidol.
4. Enhanced serotonergic effects may result from combined therapy of highly selective serotonin re-uptake inhibitors, such as fluoxetine and fluvoxamine, with lithium. Although such an enhancement may be beneficial in some instances it can produce a life-threatening serotonin syndrome comprising hyperthermia, tremor and convulsion; although the addition of lithium to tricyclic antidepressant therapy is often performed in patients with refractory depression, epileptic seizures have been reported in a patient receiving amitriptyline when lithium was added.
5. SGLT2 inhibitors may increase renal lithium excretion and the blood lithium levels may be decreased. Serum concentration of lithium should be monitored more frequently after initiation and dose changes of SGLT2 inhibitors.

PREGNANCY AND LACTATION:

Neonates exposed to antipsychotic drugs during the third trimester of pregnancy are at risk for extrapyramidal and/or withdrawal symptoms following delivery. There have been reports of agitation, hypertonia, hypotonia, tremor, somnolence, respiratory distress, and feeding disorder in these neonates. These complications have varied in severity; while in some cases symptoms have been self-limited, in other cases neonates have required intensive care unit support and prolonged hospitalization. LICONATE TABLETS 300MG should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.

SIDE EFFECTS:

Drug reaction with eosinophilia and systemic symptoms (DRESS) cases have been reported with the use of lithium.

Adverse effects occurring at the therapeutic serum-lithium concentrations include anorexia, constipation or diarrhoea, epigastric discomfort, metallic taste, headache, vertigo, fine tremor, polyuria with polydipsia, oedema (which should not be treated with diuretics), muscle weakness, feeling of somnolence and lethargy, hypermagnesaemia, cardiac arrhythmias with ECG changes, and exacerbation of skin conditions such as acne and psoriasis, leucocytosis is relatively common. Some of these effects often abate with continued therapy.

Significant weight gain may occur. Long term adverse effects include hypothyroidism and/or goitre, rarely hyperthyroidism, and mild cognitive and memory impairment.

Toxic effects may be expected at serum-lithium concentrations of about 1.5mmol/L, although they can appear at lower concentrations. They call immediate withdrawal of treatment and should always be considered very seriously. Signs of toxicity include increasing diarrhoea, vomiting, anorexia, polyuria, muscle weakness, lethargy, ataxia, lack of co-ordination, tinnitus, blurred vision, coarse tremor of the extremities and lower jaw, muscle hyper-irritability and twitching, hyper-reflexia, choreoathetoid movements, dysarthria, disorientation, psychosis, drowsiness, and seizures, coma and death may occur.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Symptoms : Nausea, vomiting and diarrhoea are common features and are followed by coarse tremor, increase muscle tone, cogwheel rigidity, fasciculation, and myoclonus. Coma and convulsions could occur in the most serious cases and cardiac effects (first degree heart block and QRS and QT prolongation) have been described rarely. Acute renal failure

and nephrogenic diabetes insipidus may develop.

Treatment : In the case of recent acute overdosage the stomach should be emptied by either induction of emesis or by lavage. Both procedures should be performed within 2-3 hours if ingestion. The use of activated charcoal and cathartics following emesis and lavage is probably not beneficial. Further measure involve procedures aimed at enhancing the renal clearance of lithium or its active removal.

Lithium poisoning may also develop during the course of therapeutic lithium administration. In some instances, temporarily withdrawal of lithium therapy may be all that is required, but in any serious or severe case of intoxication active measures such as forced diuresis or dialysis may need to be instituted.

The protocol for forced diuresis should be the relatively rapid intravenous infusion of physiological sodium citrate (for example 1 – 2 litres given over 6 hours). However, fluid balance and serum electrolytes must be strictly monitored. Dialysis is of greatest benefit when the serum concentrations are high. The recommended duration of haemodialysis is 8 to 12 hours and may be repeated based upon the patient's clinical condition and serum-lithium concentration. Haemodialysis is much more efficient in removing lithium than peritoneal dialysis and is the recommended method of choice when dialysis is indicated.

EFFECT ON ABILITY TO DRIVE AND USE MACHINE:

Since lithium may impair mental and/or physical abilities, patients should be cautioned about undertaking activities requiring alertness (e.g. operating vehicles or machinery).

PRECLINICAL SAFETY DATA:

Not applicable

INSTRUCTION FOR USE:

For oral use only.

STORAGE CONDITIONS:

Store below 30°C in well-closed containers. Avoid heat and moisture.

Keep out of reach of children / Jauhi dari kanak-kanak.

DOSAGE FORMS AND PACKAGING AVAILABLE:

Pack size: Blisterpack : 100 x 10's

NAME AND ADDRESS OF MANUFACTURER / PRODUCT REGISTRATION HOLDER:

MALAYSIAN PHARMACEUTICAL INDUSTRIES SDN BHD (101323-U)

Plot 14, Lebuhraya Kampung Jawa, 11900 Bayan Lepas,
Pulau Pinang, Malaysia.

DATE OF REVISION:

27/11/2025